

AP_Assignment_Vu Ngoc Thien Kim_22245448

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Introduction

This report presents the analysis of *FIFA World Cup match data* using the **R** programming language and the **tidyverse** package in **RStudio**.

The assignment aims to demonstrate the following **data analysis skills**:

- **Importing and preparing datasets**
- **Exploring variables**
- **Summarising information**
- **Creating appropriate data visualisations**

Four questions are completed using the FIFA World Cup datasets to investigate **match statistics** and **team performance**.

In addition, **Git** and **GitHub** are used for **version control**, allowing the development process to be tracked through meaningful commits.

The report presents the **R code**, **analysis**, **visualisations**, and **results** for each question in a clear, logical, and well-structured format.

Question 1 – Data Inspection, Cleaning and Visualisation

Objective

Write the code to inspect the data structure and present the data: The missing values in the dataset were written as “?”, replace any “?” with NA; Convert categorical variables Result, Stage, Country, and Extratime to factors; Convert variables HomePenalty and AwayPenalty to numeric format and replace the missing values with 0; Select an appropriate chart type to visualize the distribution of HomePenalty for matches that involved a penalty shootout. Briefly justify your choice and interpret the resulting plot.

R Code

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.6.1
```

```
## Warning: package 'ggplot2' was built under R version 4.6.1
```

```
## Warning: package 'tibble' was built under R version 4.6.1
```

```
## Warning: package 'tidyr' was built under R version 4.6.1
```

```
## Warning: package 'readr' was built under R version 4.6.1
```

```
## Warning: package 'purrr' was built under R version 4.6.1
```

```
## Warning: package 'dplyr' was built under R version 4.6.1
```

```
## Warning: package 'stringr' was built under R version 4.6.1
```

```
## Warning: package 'forcats' was built under R version 4.6.1
```

```
## Warning: package 'lubridate' was built under R version 4.6.1
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.3      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
#####
# PART B - Import and Inspect the datasets
#####

# Read the Matches dataset.
# This dataset contains information about every FIFA World Cup
# match.

matches <-
  read.csv(
    "data/Matches.csv",
    stringsAsFactors = FALSE
  )

# Read the Teams dataset.
# This dataset stores the country name and FIFA team code.

teams <-
  read.csv(
    "data/Teams.csv",
    stringsAsFactors = FALSE
  )

# Read the Stadiums dataset.
# This dataset stores stadium locations.

stadiums <-
  read.csv(
    "data/Stadiums.csv",
    stringsAsFactors = FALSE
  )

# Read the Tournament dataset.
# This dataset contains tournament editions.

tournaments <-
  read.csv(
    "data/Tournaments.csv",
    stringsAsFactors = FALSE
  )

#####
# Inspect the datasets
#####

# Display the first six rows to understand the structure.

head(matches)
```

##	MatchID	TournamentID	MatchName	Stage	MatchDate
## 1	M-1930-01	WC-1930	France v Mexico	group stage	7/13/1930
## 2	M-1930-02	WC-1930	United States v Belgium	group stage	7/13/1930
## 3	M-1930-03	WC-1930	Yugoslavia v Brazil	group stage	7/14/1930

```
## 4 M-1930-04      WC-1930      Romania v Peru group stage 7/14/1930
## 5 M-1930-05      WC-1930      Argentina v France group stage 7/15/1930
## 6 M-1930-06      WC-1930      Chile v Mexico group stage 7/16/1930
##   MatchTime StadiumID HomeTeamID AwayTeamID HomeTeamScore AwayTeamScore
## 1  15:00:00    S-193      T-28      T-44              4              1
## 2  15:00:00    S-192      T-80      T-06              3              0
## 3  12:45:00    S-192      T-84      T-09              2              1
## 4  14:50:00    S-193      T-59      T-54              3              1
## 5  16:00:00    S-192      T-03      T-28              1              0
## 6  14:45:00    S-192      T-13      T-44              3              0
##   ExtraTime PenaltyShootout HomePenalty AwayPenalty      Result
## 1         0              0          ?          0 home team win
## 2         0              0          0          0 home team win
## 3         0              0          0          0 home team win
## 4         0              0          0          ? home team win
## 5         0              0          0          0 home team win
## 6         0              0          0          0 home team win
```

```
head(teams)
```

```
##   TeamID TeamName TeamCode
## 1   T-01   Algeria      DZA
## 2   T-02    Angola      AGO
## 3   T-03 Argentina      ARG
## 4   T-04 Australia      AUS
## 5   T-05   Austria      AUT
## 6   T-06   Belgium      BEL
```

```
head(stadiums)
```

```
##   StadiumID      StadiumName      City Country
## 1     S-089      Jita Stadium      Jita   Japan
## 2     S-108 Ahmad bin Ali Stadium Al Rayyan Qatar
## 3     S-107      Al Bayt Stadium Al Khor   Qatar
## 4     S-111      Al Janoub Stadium Al Wakrah Qatar
## 5     S-112      Al Thumama Stadium Doha    Qatar
## 6     S-064      Allianz Arena      Munich Germany
```

```
head(tournaments)
```

```
##   TournamentID      TournamentName
## 1      WC-1930 1930 FIFA World Cup
## 2      WC-1934 1934 FIFA World Cup
## 3      WC-1938 1938 FIFA World Cup
## 4      WC-1950 1950 FIFA World Cup
## 5      WC-1954 1954 FIFA World Cup
## 6      WC-1958 1958 FIFA World Cup
```

```
# Display the structure of each dataset.
```

```
str(matches)
```

```
## 'data.frame':   964 obs. of  16 variables:
## $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName     : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v" ...
## $ Stage         : chr  "group stage" "group stage" "group stage" "group stage" ...
## $ MatchDate     : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime     : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
## $ StadiumID     : chr  "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID    : chr  "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID    : chr  "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore : int  4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore : int  1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime     : int  0 0 0 0 0 0 0 0 0 0 ...
## $ PenaltyShootout: int  0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty   : chr  "?" "0" "0" "0" ...
## $ AwayPenalty   : chr  "0" "0" "0" "?" ...
## $ Result        : chr  "home team win" "home team win" "home team win" "home team win" ...
```

```
str(teams)
```

```
## 'data.frame':   85 obs. of  3 variables:
## $ TeamID      : chr  "T-01" "T-02" "T-03" "T-04" ...
## $ TeamName    : chr  "Algeria" "Angola" "Argentina" "Australia" ...
## $ TeamCode    : chr  "DZA" "AGO" "ARG" "AUS" ...
```

```
str(stadiums)
```

```
## 'data.frame':   193 obs. of  4 variables:
## $ StadiumID   : chr  "S-089" "S-108" "S-107" "S-111" ...
## $ StadiumName : chr  "Jita Stadium" "Ahmad bin Ali Stadium" "Al Bayt Stadium" "Al Janoub Stadium" ..
## $ City        : chr  "Jita" "Al Rayyan" "Al Khor" "Al Wakrah" ...
## $ Country     : chr  "Japan" "Qatar" "Qatar" "Qatar" ...
```

```
str(tournaments)
```

```
## 'data.frame':   23 obs. of  2 variables:
## $ TournamentID : chr  "WC-1930" "WC-1934" "WC-1938" "WC-1950" ...
## $ TournamentName: chr  "1930 FIFA World Cup" "1934 FIFA World Cup" "1938 FIFA World Cup" "1950 FIFA World Cup" ...
```

```
# Display summary statistics and detect possible missing values.
```

```
summary(matches)
```

```
##      MatchID      TournamentID      MatchName      Stage
## Length :964 Length :964 Length :964 Length :964
## N.unique :964 N.unique : 22 N.unique :759 N.unique : 8
## N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 9 Min.nchar: 7 Min.nchar: 11 Min.nchar: 5
## Max.nchar: 9 Max.nchar: 7 Max.nchar: 35 Max.nchar: 18
##
##      MatchDate      MatchTime      StadiumID      HomeTeamID
```

```
## Length :964 Length :964 Length :964 Length :964
## N.unique :378 N.unique : 35 N.unique :193 N.unique : 81
## N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 9 Min.nchar: 8 Min.nchar: 5 Min.nchar: 4
## Max.nchar: 10 Max.nchar: 8 Max.nchar: 5 Max.nchar: 4
##
## AwayTeamID HomeTeamScore AwayTeamScore ExtraTime
## Length :964 Min. : 0.000 Min. :0.000 Min. :0.00000
## N.unique : 85 1st Qu.: 1.000 1st Qu.:0.000 1st Qu.:0.00000
## N.blank : 0 Median : 1.000 Median :1.000 Median :0.00000
## Min.nchar: 4 Mean : 1.767 Mean :1.055 Mean :0.07573
## Max.nchar: 4 3rd Qu.: 3.000 3rd Qu.:2.000 3rd Qu.:0.00000
## Max. :10.000 Max. :7.000 Max. :1.00000
## PenaltyShootout HomePenalty AwayPenalty Result
## Min. :0.00000 Length :964 Length :964 Length :964
## 1st Qu.:0.00000 N.unique : 7 N.unique : 7 N.unique : 3
## Median :0.00000 N.blank : 0 N.blank : 0 N.blank : 0
## Mean :0.03631 Min.nchar: 1 Min.nchar: 1 Min.nchar: 4
## 3rd Qu.:0.00000 Max.nchar: 1 Max.nchar: 1 Max.nchar: 13
## Max. :1.00000
```

```
summary(teams)
```

```
## TeamID TeamName TeamCode
## Length :85 Length :85 Length :85
## N.unique :85 N.unique :85 N.unique :84
## N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 4 Min.nchar: 4 Min.nchar: 3
## Max.nchar: 4 Max.nchar:22 Max.nchar: 3
```

```
summary(stadiums)
```

```
## StadiumID StadiumName City Country
## Length :193 Length :193 Length :193 Length :193
## N.unique :193 N.unique :192 N.unique :165 N.unique : 18
## N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 5 Min.nchar: 6 Min.nchar: 4 Min.nchar: 5
## Max.nchar: 5 Max.nchar: 30 Max.nchar: 16 Max.nchar: 13
```

```
summary(tournaments)
```

```
## TournamentID TournamentName
## Length :23 Length :23
## N.unique :23 N.unique :23
## N.blank : 0 N.blank : 0
## Min.nchar: 7 Min.nchar:19
## Max.nchar: 7 Max.nchar:19
```

```
#####
# PART C - Clean the data
#####
```

```
#####
# Step 1: Check missing values represented by "?"
#####

# Count the number of "?" values in each dataset.

# Count the number of "?" values in the Matches dataset.
sum(matches == "?", na.rm = TRUE)
```

```
## [1] 67
```

```
# Count the number of "?" values in the Stadiums dataset.
sum(stadiums == "?", na.rm = TRUE)
```

```
## [1] 0
```

```
#####
# Step 2: Replace "?" with NA
#####

# Replace all occurrences of "?" with NA.
# NA is the standard representation of missing values in R.
# Replace all "?" values with NA so that R recognises
# them as missing values.

matches[matches == "?"] <- NA

stadiums[stadiums == "?"] <- NA

# Verify that all "?" values have been removed.

sum(matches == "?", na.rm = TRUE)
```

```
## [1] 0
```

```
sum(stadiums == "?", na.rm = TRUE)
```

```
## [1] 0
```

```
#####
# Step 3: Convert categorical variables into factors
#####

# Result contains categorical values.

matches$Result <-
  factor(matches$Result)

# Stage contains competition stages.
```



```
matches$Stage <-
  factor(matches$Stage)
```

ExtraTime contains 2 categorical values (0 and 1).

```
matches$ExtraTime <-
  factor(matches$ExtraTime)
```

Country is a categorical variable.

```
stadiums$Country <-
  factor(stadiums$Country)
```

Check the updated data types.

```
str(matches)
```

```
## 'data.frame':   964 obs. of  16 variables:
##  $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
##  $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
##  $ MatchName     : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v Hungary" ...
##  $ Stage         : Factor w/ 8 levels "final","final round",...: 3 3 3 3 3 3 3 3 ...
##  $ MatchDate     : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
##  $ MatchTime     : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
##  $ StadiumID     : chr  "S-193" "S-192" "S-192" "S-193" ...
##  $ HomeTeamID    : chr  "T-28" "T-80" "T-84" "T-59" ...
##  $ AwayTeamID    : chr  "T-44" "T-06" "T-09" "T-54" ...
##  $ HomeTeamScore  : int   4 3 2 3 1 3 4 3 1 1 ...
##  $ AwayTeamScore  : int   1 0 1 1 0 0 0 0 0 0 ...
##  $ ExtraTime      : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 ...
##  $ PenaltyShootout: int   0 0 0 0 0 0 0 0 0 0 ...
##  $ HomePenalty    : chr   NA "0" "0" "0" ...
##  $ AwayPenalty    : chr   "0" "0" "0" NA ...
##  $ Result         : Factor w/ 3 levels "away team win",...: 3 3 3 3 3 3 3 3 3 ...
```

```
str(stadiums)
```

```
## 'data.frame':   193 obs. of  4 variables:
##  $ StadiumID    : chr  "S-089" "S-108" "S-107" "S-111" ...
##  $ StadiumName   : chr  "Jita Stadium" "Ahmad bin Ali Stadium" "Al Bayt Stadium" "Al Janoub Stadium" ...
##  $ City          : chr  "Jita" "Al Rayyan" "Al Khor" "Al Wakrah" ...
##  $ Country       : Factor w/ 18 levels "Argentina","Brazil",...: 8 10 10 10 10 6 6 2 2 2 ...
```

```
#####
# Step 4: Convert penalty variables into numeric
#####
```

Convert HomePenalty into numeric.

```
matches$HomePenalty <-
  as.numeric(matches$HomePenalty)
```

```
# Convert AwayPenalty into numeric.
```

```
matches$AwayPenalty <-  
  as.numeric(matches$AwayPenalty)
```

```
#####  
# Step 5: Replace missing penalty values with zero  
#####
```

```
# Missing values indicate no penalty goals.
```

```
matches$HomePenalty[  
  is.na(matches$HomePenalty)  
] <- 0
```

```
matches$AwayPenalty[  
  is.na(matches$AwayPenalty)  
] <- 0
```

```
#####  
# Step 6: Check the cleaned data  
#####  
str(matches)
```

```
## 'data.frame':   964 obs. of  16 variables:  
##  $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...  
##  $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...  
##  $ MatchName     : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v  
##  $ Stage        : Factor w/ 8 levels "final","final round",...: 3 3 3 3 3 3 3 3 3 ...  
##  $ MatchDate     : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...  
##  $ MatchTime     : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...  
##  $ StadiumID     : chr  "S-193" "S-192" "S-192" "S-193" ...  
##  $ HomeTeamID    : chr  "T-28" "T-80" "T-84" "T-59" ...  
##  $ AwayTeamID    : chr  "T-44" "T-06" "T-09" "T-54" ...  
##  $ HomeTeamScore : int   4 3 2 3 1 3 4 3 1 1 ...  
##  $ AwayTeamScore : int   1 0 1 1 0 0 0 0 0 0 ...  
##  $ ExtraTime     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 ...  
##  $ PenaltyShootout: int   0 0 0 0 0 0 0 0 0 ...  
##  $ HomePenalty   : num   0 0 0 0 0 0 0 0 0 ...  
##  $ AwayPenalty   : num   0 0 0 0 0 0 0 0 0 ...  
##  $ Result        : Factor w/ 3 levels "away team win",...: 3 3 3 3 3 3 3 3 3 ...
```

```
summary(matches$HomePenalty)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   
## 0.0000  0.0000  0.0000  0.1048  0.0000  5.0000
```

```
summary(matches$AwayPenalty)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   
## 0.0000  0.0000  0.0000  0.1089  0.0000  5.0000
```

```
sum(is.na(matches$HomePenalty))
```

```
## [1] 0
```

```
sum(is.na(matches$AwayPenalty))
```

```
## [1] 0
```

```
#####  
# PART D - Prepare the data for visualisation  
#####  
  
#####  
# Check values of PenaltyShootout  
#####  
  
# Display the number of matches with and without  
# a penalty shootout.  
  
table(matches$PenaltyShootout)
```

```
##  
##    0    1  
## 929  35
```

```
# The value 1 indicates that the match  
# was decided by a penalty shootout.  
  
#####  
# Filter matches decided by penalty shootout  
#####  
  
# Filter the dataset to include only matches  
# decided by a penalty shootout.  
# Store the filtered data in a new object  
# for plotting.  
penalty_matches <-  
  matches %>%  
  filter(PenaltyShootout == 1)  
  
#####  
# PART E - Create the visualisation  
#####  
  
# A histogram is used because HomePenalty is a  
# numerical variable and the chart clearly shows  
# the distribution of penalty goals.  
  
ggplot(  
  penalty_matches,
```

```

aes(
  x = HomePenalty
) +
geom_histogram(
  binwidth = 1,
  colour = "black",
  fill = "steelblue"
) +
labs(
  title = "Distribution of Home Penalty Goals",
  x = "Home Penalty Goals",
  y = "Number of Matches"
) +
theme_minimal()

```

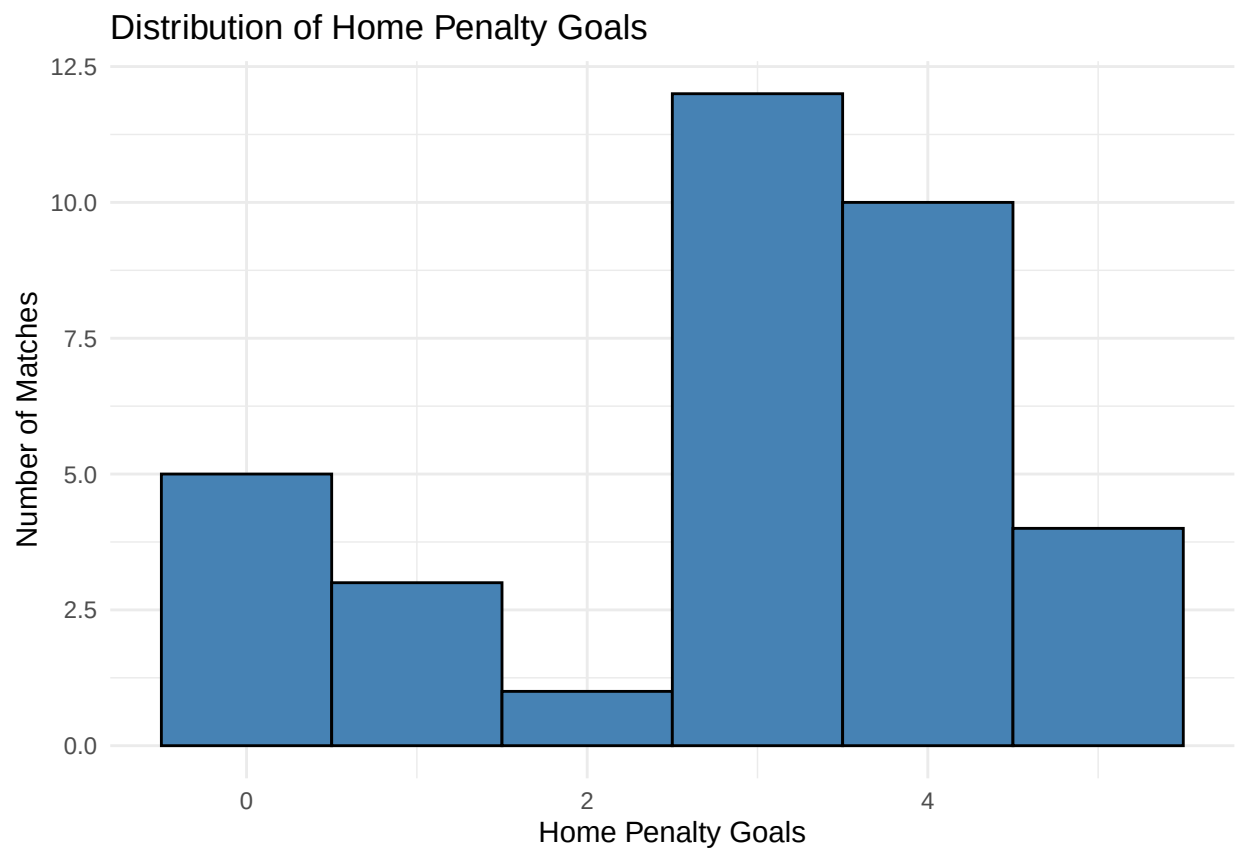


Figure 1. Distribution of Home Penalty Goals.

```

#####
# PART F - Validate the results
#####

# Check data types

str(matches)

```

```
## 'data.frame':    964 obs. of  16 variables:
## $ MatchID       : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID  : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName     : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v ..."
## $ Stage         : Factor w/ 8 levels "final","final round",...: 3 3 3 3 3 3 3 3 ...
## $ MatchDate     : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime     : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
## $ StadiumID     : chr  "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID    : chr  "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID    : chr  "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore  : int   4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore  : int   1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime      : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 ...
## $ PenaltyShootout: int   0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty    : num   0 0 0 0 0 0 0 0 0 0 ...
## $ AwayPenalty    : num   0 0 0 0 0 0 0 0 0 0 ...
## $ Result         : Factor w/ 3 levels "away team win",...: 3 3 3 3 3 3 3 3 3 ...
```

```
# Check summary statistics
```

```
summary(matches)
```

```
##      MatchID      TournamentID      MatchName      Stage
## Length :964 Length :964 Length :964 group stage :676
## N.unique :964 N.unique : 22 N.unique :759 round of 16 : 97
## N.blank : 0 N.blank : 0 N.blank : 0 quarter-finals : 70
## Min.nchar: 9 Min.nchar: 7 Min.nchar: 11 semi-finals : 38
## Max.nchar: 9 Max.nchar: 7 Max.nchar: 35 second group stage: 36
## final : 21
## (Other) : 26
##      MatchDate      MatchTime      StadiumID      HomeTeamID
## Length :964 Length :964 Length :964 Length :964
## N.unique :378 N.unique : 35 N.unique :193 N.unique : 81
## N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 9 Min.nchar: 8 Min.nchar: 5 Min.nchar: 4
## Max.nchar: 10 Max.nchar: 8 Max.nchar: 5 Max.nchar: 4
##
##      AwayTeamID HomeTeamScore AwayTeamScore ExtraTime PenaltyShootout
## Length :964 Min. : 0.000 Min. :0.000 0:891 Min. :0.00000
## N.unique : 85 1st Qu.: 1.000 1st Qu.:0.000 1: 73 1st Qu.:0.00000
## N.blank : 0 Median : 1.000 Median :1.000 Median :0.00000
## Min.nchar: 4 Mean : 1.767 Mean :1.055 Mean :0.03631
## Max.nchar: 4 3rd Qu.: 3.000 3rd Qu.:2.000 3rd Qu.:0.00000
## Max. :10.000 Max. :7.000 Max. :1.00000
##
##      HomePenalty      AwayPenalty      Result
## Min. :0.0000 Min. :0.0000 away team win:240
## 1st Qu.:0.0000 1st Qu.:0.0000 draw :179
## Median :0.0000 Median :0.0000 home team win:545
## Mean :0.1048 Mean :0.1089
## 3rd Qu.:0.0000 3rd Qu.:0.0000
## Max. :5.0000 Max. :5.0000
##
```

```
#Check penalty values
```

```
summary(matches$HomePenalty)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1048  0.0000  5.0000
```

```
summary(matches$AwayPenalty)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1089  0.0000  5.0000
```

```
# Check filtered data
```

```
head(penalty_matches)
```

```
##      MatchID TournamentID      MatchName      Stage
## 1 M-1982-50      WC-1982    West Germany v France semi-finals
## 2 M-1986-45      WC-1986      Brazil v France quarter-finals
## 3 M-1986-46      WC-1986    West Germany v Mexico quarter-finals
## 4 M-1986-48      WC-1986      Spain v Belgium quarter-finals
## 5 M-1990-41      WC-1990 Republic of Ireland v Romania round of 16
## 6 M-1990-45      WC-1990    Argentina v Yugoslavia quarter-finals
##      MatchDate MatchTime StadiumID HomeTeamID AwayTeamID HomeTeamScore
## 1 07/08/1982  21:00:00    S-159      T-83      T-28          3
## 2  6/21/1986  12:00:00    S-095      T-09      T-28          1
## 3  6/21/1986  16:00:00    S-102      T-83      T-44          0
## 4  6/22/1986  16:00:00    S-104      T-71      T-06          1
## 5  6/25/1990  17:00:00    S-073      T-58      T-59          0
## 6  6/30/1990  17:00:00    S-071      T-03      T-84          0
##      AwayTeamScore ExtraTime PenaltyShootout HomePenalty AwayPenalty      Result
## 1          3          1          1          0          4 home team win
## 2          1          1          1          0          4 away team win
## 3          0          1          1          0          1 home team win
## 4          1          1          1          0          5 away team win
## 5          0          1          1          5          4 home team win
## 6          0          1          1          3          2 home team win
```

```
getwd()
```

```
## [1] "C:/Users/User/OneDrive/Tài liệu/COMP1013_Project"
```

Question 2 – Home Team Score Analysis

Objective

Write the code to analyse the distribution of HomeTeamScore across different competition stages (Stage) and tournament name (TournamentName). Then, write the code to investigate the distribution of HomeTeamScore across the following groups of AwayTeamScore (e.g., 0-1 goals, 2-3 goals, 4 or more goals). Visualise the findings using histograms and briefly explain your observations.

R Code

Data Preparation

```
# PART A - Load required packages
#####
library(tidyverse)

#####
# PART B - Import and prepare the datasets
#####
# Read the Matches dataset.
# This dataset contains match scores and competition stages.

matches <-
  read.csv(
    "data/Matches.csv",
    stringsAsFactors = FALSE
  )

# Read the Tournament dataset.
# This dataset contains tournament names.

tournaments <-
  read.csv(
    "data/Tournaments.csv",
    stringsAsFactors = FALSE
  )

#####
# Merge datasets
#####
# Merge Matches with Tournaments to obtain the tournament name.

matches_full <-
  matches %>%
  left_join(
    tournaments,
    by = "TournamentID"
  )

#####
# Check merged dataset
```

```
# Display the first six rows.
```

```
head(matches_full)
```

```
##      MatchID TournamentID      MatchName      Stage MatchDate
## 1 M-1930-01      WC-1930      France v Mexico group stage 7/13/1930
## 2 M-1930-02      WC-1930 United States v Belgium group stage 7/13/1930
## 3 M-1930-03      WC-1930      Yugoslavia v Brazil group stage 7/14/1930
## 4 M-1930-04      WC-1930      Romania v Peru group stage 7/14/1930
## 5 M-1930-05      WC-1930      Argentina v France group stage 7/15/1930
## 6 M-1930-06      WC-1930      Chile v Mexico group stage 7/16/1930
##      MatchTime StadiumID HomeTeamID AwayTeamID HomeTeamScore AwayTeamScore
## 1 15:00:00      S-193      T-28      T-44              4              1
## 2 15:00:00      S-192      T-80      T-06              3              0
## 3 12:45:00      S-192      T-84      T-09              2              1
## 4 14:50:00      S-193      T-59      T-54              3              1
## 5 16:00:00      S-192      T-03      T-28              1              0
## 6 14:45:00      S-192      T-13      T-44              3              0
##      ExtraTime PenaltyShootout HomePenalty AwayPenalty      Result
## 1          0              0          ?          0 home team win
## 2          0              0          0          0 home team win
## 3          0              0          0          0 home team win
## 4          0              0          0          ? home team win
## 5          0              0          0          0 home team win
## 6          0              0          0          0 home team win
##      TournamentName
## 1 1930 FIFA World Cup
## 2 1930 FIFA World Cup
## 3 1930 FIFA World Cup
## 4 1930 FIFA World Cup
## 5 1930 FIFA World Cup
## 6 1930 FIFA World Cup
```

```
# Display the data structure.
```

```
str(matches_full)
```

```
## 'data.frame':    964 obs. of  17 variables:
## $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName     : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v" ...
## $ Stage         : chr  "group stage" "group stage" "group stage" "group stage" ...
## $ MatchDate     : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime     : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
## $ StadiumID     : chr  "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID    : chr  "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID    : chr  "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore : int   4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore : int   1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime     : int   0 0 0 0 0 0 0 0 0 0 ...
## $ PenaltyShootout: int   0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty   : chr   "?" "0" "0" "0" ...
## $ AwayPenalty   : chr   "0" "0" "0" "?" ...
## $ Result        : chr  "home team win" "home team win" "home team win" "home team win" ...
```



```
## $ TournamentName : chr "1930 FIFA World Cup" "1930 FIFA World Cup" "1930 FIFA World Cup" "1930 FIF
```

```
# Convert Stage and TournamentName into factors.  
# Stage is a categorical variable.
```

```
matches_full$Stage <-  
  factor(matches_full$Stage)
```

```
# TournamentName is a categorical variable.
```

```
matches_full$TournamentName <-  
  factor(matches_full$TournamentName)
```

```
# Check the updated data types.
```

```
str(matches_full)
```

```
## 'data.frame': 964 obs. of 17 variables:  
## $ MatchID : chr "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...  
## $ TournamentID : chr "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...  
## $ MatchName : chr "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v ...  
## $ Stage : Factor w/ 8 levels "final","final round",...: 3 3 3 3 3 3 3 3 3 3 ...  
## $ MatchDate : chr "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...  
## $ MatchTime : chr "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...  
## $ StadiumID : chr "S-193" "S-192" "S-192" "S-193" ...  
## $ HomeTeamID : chr "T-28" "T-80" "T-84" "T-59" ...  
## $ AwayTeamID : chr "T-44" "T-06" "T-09" "T-54" ...  
## $ HomeTeamScore : int 4 3 2 3 1 3 4 3 1 1 ...  
## $ AwayTeamScore : int 1 0 1 1 0 0 0 0 0 0 ...  
## $ ExtraTime : int 0 0 0 0 0 0 0 0 0 0 ...  
## $ PenaltyShootout: int 0 0 0 0 0 0 0 0 0 0 ...  
## $ HomePenalty : chr "?" "0" "0" "0" ...  
## $ AwayPenalty : chr "0" "0" "0" "?" ...  
## $ Result : chr "home team win" "home team win" "home team win" "home team win" ...  
## $ TournamentName : Factor w/ 22 levels "1930 FIFA World Cup",...: 1 1 1 1 1 1 1 1 1 1 ...
```

Distribution of Home Team Score by Competition Stage

```
# PART C - Analyse HomeTeamScore by Stage and TournamentName
#####

# Check Stage values
#####

# Display the frequency of each competition stage.

table(matches_full$Stage)
```

```
##
##           final           final round           group stage           quarter-finals
##           21             6             676             70
##   round of 16 second group stage           semi-finals third-place match
##           97             36             38             20
```

```
#####
# Create a histogram of HomeTeamScore by Stage
#####
# A histogram is used because HomeTeamScore is a numerical variable
# and the chart clearly shows its distribution.
```

```
ggplot(
  matches_full,
  aes(
    x = HomeTeamScore
  )
) +
  geom_histogram(
    binwidth = 1,
    colour = "black",
    fill = "steelblue"
  ) +
  facet_wrap(
    ~ Stage,
    ncol = 4,
    labeller = labeller(
      Stage = c(
        "group stage" = "Group Stage",
        "second group stage" = "Second Group Stage",
        "round of 16" = "Round of 16",
        "quarter-finals" = "Quarter-finals",
        "semi-finals" = "Semi-finals",
        "third-place match" = "Third-place Match",
        "final" = "Final",
        "final round" = "Final Round"
      )
    )
  ) +
  labs(
```

```

title = "Distribution of Home Team Scores by Competition Stage",
x = "Home Team Score",
y = "Number of Matches"
) +
theme_minimal() +
theme(
  plot.title = element_text(
    face = "bold",
    hjust = 0.5
  ),
  strip.text = element_text(
    face = "bold"
  ),
  panel.grid.minor = element_blank()
) +
scale_x_continuous(
  breaks = 0:10
) +
scale_y_continuous(
  breaks = scales::pretty_breaks()
)

```

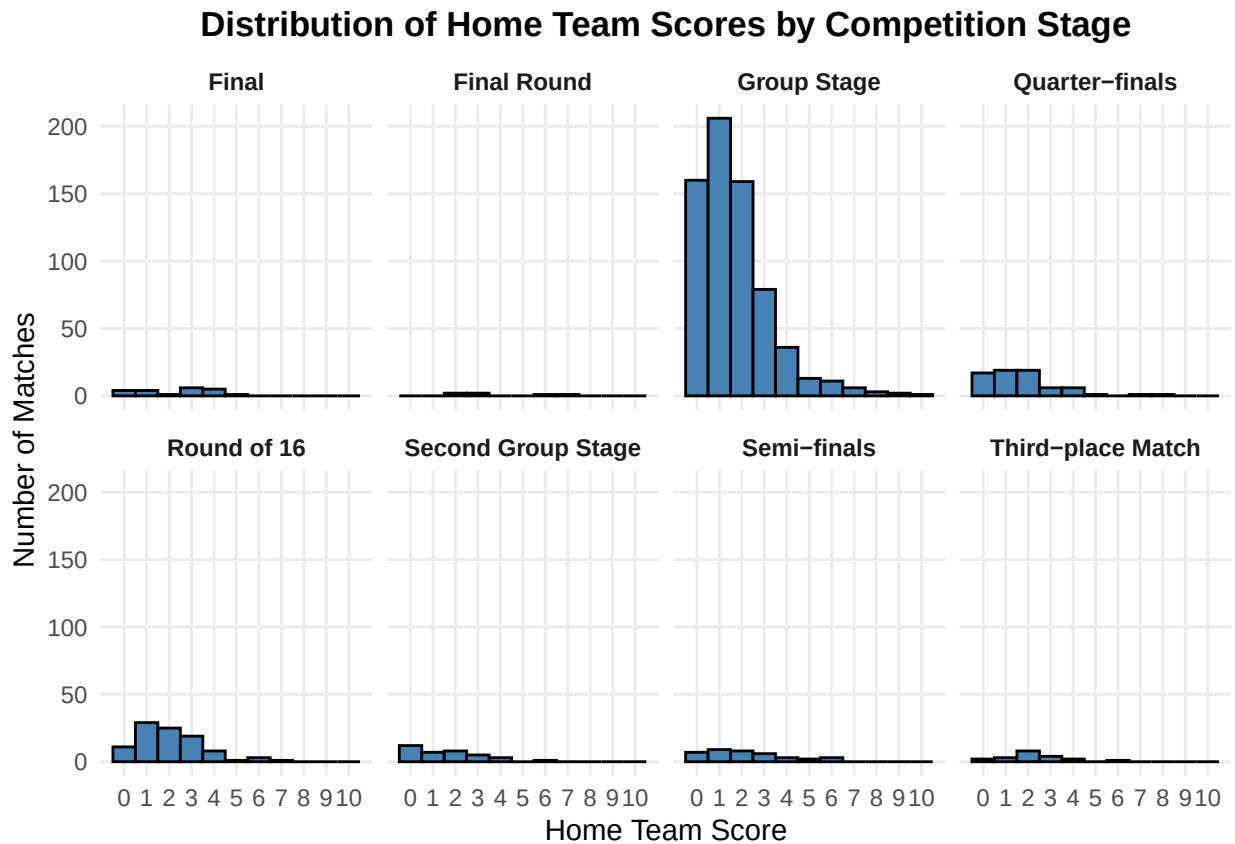


Figure 2. Distribution of Home Team Scores by Competition Stage.

Distribution of Home Team Scores by Tournament

```
# Check TournamentName values
#####

# Display the frequency of each tournament.

table(matches_full$TournamentName)

##
## 1930 FIFA World Cup 1934 FIFA World Cup 1938 FIFA World Cup 1950 FIFA World Cup
##                18                17                18                22
## 1954 FIFA World Cup 1958 FIFA World Cup 1962 FIFA World Cup 1966 FIFA World Cup
##                26                35                32                32
## 1970 FIFA World Cup 1974 FIFA World Cup 1978 FIFA World Cup 1982 FIFA World Cup
##                32                38                38                52
## 1986 FIFA World Cup 1990 FIFA World Cup 1994 FIFA World Cup 1998 FIFA World Cup
##                52                52                52                64
## 2002 FIFA World Cup 2006 FIFA World Cup 2010 FIFA World Cup 2014 FIFA World Cup
##                64                64                64                64
## 2018 FIFA World Cup 2022 FIFA World Cup
##                64                64

#####
# Create a histogram of HomeTeamScore by TournamentName
#####
# A histogram is used because HomeTeamScore is a numerical variable and the chart
# clearly shows its distribution for each tournament.

ggplot(
  matches_full,
  aes(
    x = HomeTeamScore
  )
) +
  geom_histogram(
    binwidth = 1,
    colour = "black",
    fill = "darkseagreen3"
  ) +
  facet_wrap(
    ~ TournamentName,
    ncol = 4
  ) +
  labs(
    title = "Distribution of Home Team Scores by Tournament",
    x = "Home Team Score",
    y = "Number of Matches"
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(face = "bold"),
```

```

plot.title = element_text(face = "bold", hjust = 0.5),
axis.text = element_text(size = 8),
panel.grid.minor = element_blank()
)

```

```

## Warning in plot_theme(plot): The `panel.grid.minor` theme element is not
## defined in the element hierarchy.

```

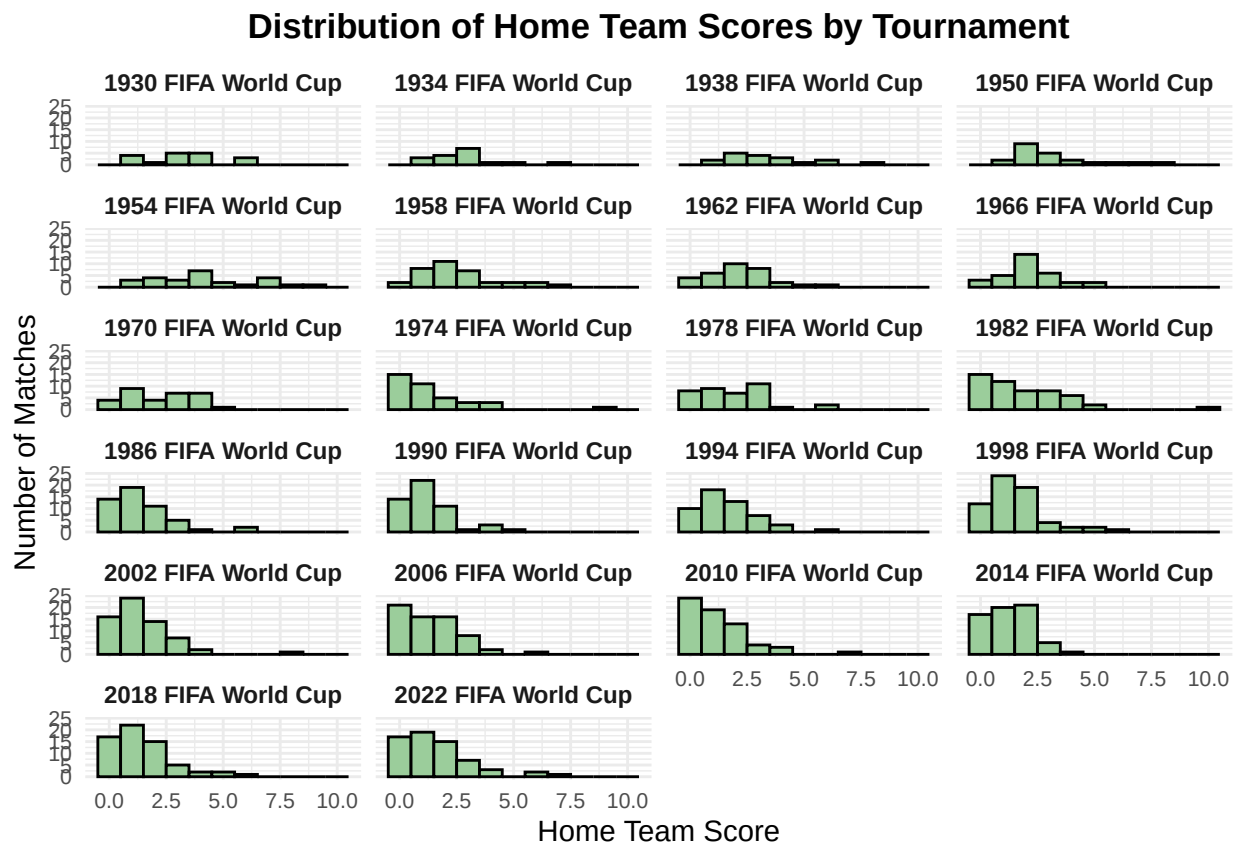


Figure 3. Distribution of Home Team Scores by Tournaments.

Distribution of Home Team Scores by Away Team Score Groups

```

# PART D - Create AwayTeamScore groups
#####

# Create a new variable to store AwayTeamScore groups.

matches_full$AwayScoreGroup <- NA

# Group matches where the away team scored 0 or 1 goal.

matches_full$AwayScoreGroup[
  matches_full$AwayTeamScore <= 1
] <- "0-1 goals"

```

```

# Group matches where the away team scored 2 or 3 goals.

matches_full$AwayScoreGroup[
  matches_full$AwayTeamScore >= 2 &
  matches_full$AwayTeamScore <= 3
] <- "2-3 goals"

# Group matches where the away team scored 4 or more goals.

matches_full$AwayScoreGroup[
  matches_full$AwayTeamScore >= 4
] <- "4+ goals"

#####
# Check AwayTeamScore groups
#####

# Display the number of matches in each AwayTeamScore group.

table(matches_full$AwayScoreGroup)

##
## 0-1 goals 2-3 goals 4+ goals
##      701      237      26

```

Figure 4. Distribution of Home Team Scores by Away Team Score Groups.

Validation

```
# PART E - Create the visualisation
#####

# Create a histogram of HomeTeamScore by AwayTeamScore groups
#####

# A histogram is used because HomeTeamScore is a numerical variable and
# the chart clearly shows the distribution across AwayTeamScore groups.

ggplot(
  matches_full,
  aes(
    x = HomeTeamScore
  )
) +
  geom_histogram(
    binwidth = 1,
    colour = "black",
    fill = "lightblue"
  ) +
  facet_wrap(
    ~ AwayScoreGroup
  ) +
  labs(
    title = "Distribution of Home Team Scores by Away Team Score Groups",
    x = "Home Team Score",
    y = "Number of Matches"
  ) +
  theme_minimal()
```

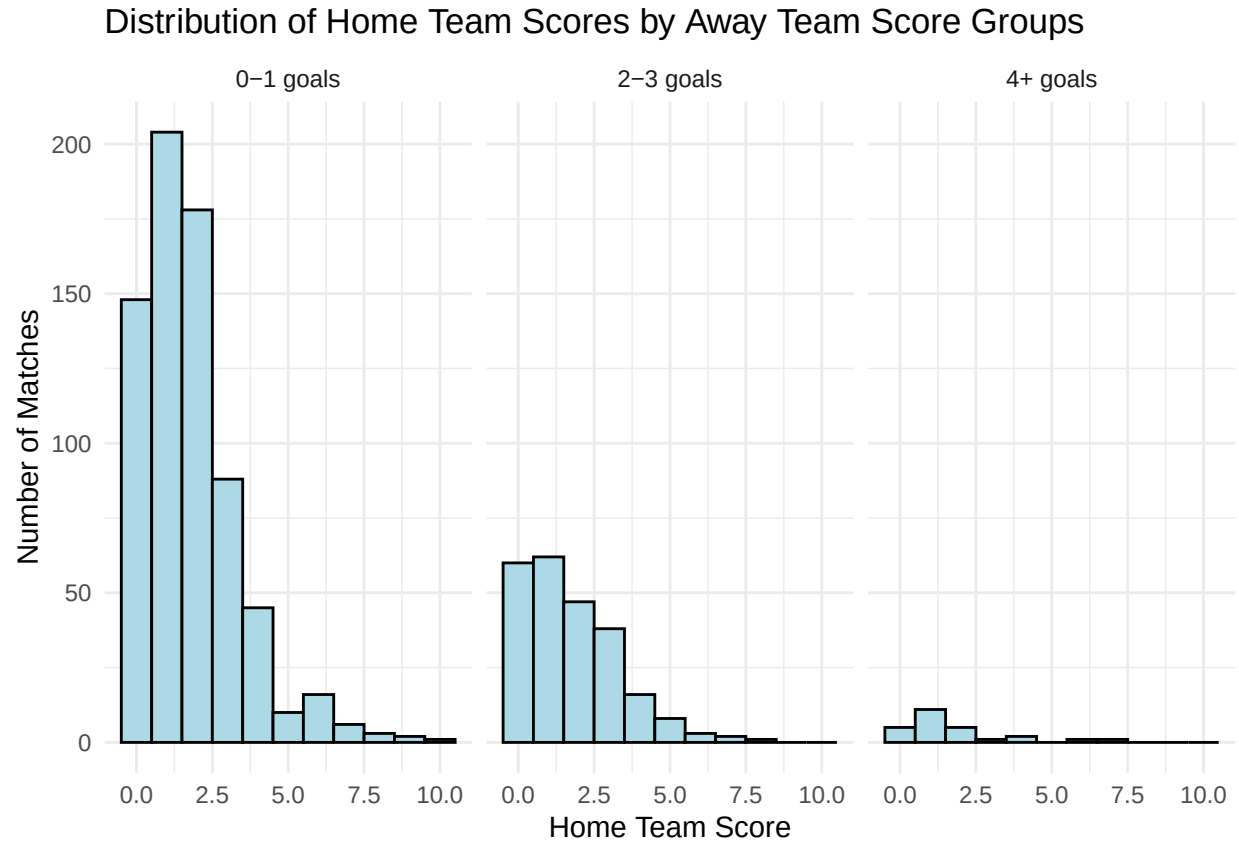


Figure 4. Distribution of Home Team Scores by Away Team Score Groups.

```
#####
# PART F - Validate the results
#####

# Check the structure of the dataset.

str(matches_full)
```

```
## 'data.frame':   964 obs. of  18 variables:
##  $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
##  $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
##  $ MatchName     : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v" ...
##  $ Stage         : Factor w/ 8 levels "final","final round",...: 3 3 3 3 3 3 3 3 3 ...
##  $ MatchDate     : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
##  $ MatchTime     : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
##  $ StadiumID     : chr  "S-193" "S-192" "S-192" "S-193" ...
##  $ HomeTeamID    : chr  "T-28" "T-80" "T-84" "T-59" ...
##  $ AwayTeamID    : chr  "T-44" "T-06" "T-09" "T-54" ...
##  $ HomeTeamScore : int   4 3 2 3 1 3 4 3 1 1 ...
##  $ AwayTeamScore : int   1 0 1 1 0 0 0 0 0 0 ...
##  $ ExtraTime     : int   0 0 0 0 0 0 0 0 0 0 ...
##  $ PenaltyShootout: int   0 0 0 0 0 0 0 0 0 0 ...
##  $ HomePenalty   : chr  "?" "0" "0" "0" ...
##  $ AwayPenalty   : chr  "0" "0" "0" "?" ...
##  $ Result        : chr  "home team win" "home team win" "home team win" "home team win" ...
```



```
## $ TournamentName : Factor w/ 22 levels "1930 FIFA World Cup",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ AwayScoreGroup : chr "0-1 goals" "0-1 goals" "0-1 goals" "0-1 goals" ...
```

```
# Check summary statistics.
```

```
summary(matches_full)
```

```
##      MatchID      TournamentID      MatchName      Stage
## Length :964 Length :964 Length :964 group stage :676
## N.unique :964 N.unique : 22 N.unique :759 round of 16 : 97
## N.blank : 0 N.blank : 0 N.blank : 0 quarter-finals : 70
## Min.nchar: 9 Min.nchar: 7 Min.nchar: 11 semi-finals : 38
## Max.nchar: 9 Max.nchar: 7 Max.nchar: 35 second group stage: 36
## final : 21
## (Other) : 26
##      MatchDate      MatchTime      StadiumID      HomeTeamID
## Length :964 Length :964 Length :964 Length :964
## N.unique :378 N.unique : 35 N.unique :193 N.unique : 81
## N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 9 Min.nchar: 8 Min.nchar: 5 Min.nchar: 4
## Max.nchar: 10 Max.nchar: 8 Max.nchar: 5 Max.nchar: 4
##
##
##      AwayTeamID      HomeTeamScore      AwayTeamScore      ExtraTime
## Length :964 Min. : 0.000 Min. :0.000 Min. :0.00000
## N.unique : 85 1st Qu.: 1.000 1st Qu.:0.000 1st Qu.:0.00000
## N.blank : 0 Median : 1.000 Median :1.000 Median :0.00000
## Min.nchar: 4 Mean : 1.767 Mean :1.055 Mean :0.07573
## Max.nchar: 4 3rd Qu.: 3.000 3rd Qu.:2.000 3rd Qu.:0.00000
## Max. :10.000 Max. :7.000 Max. :1.00000
##
##      PenaltyShootout      HomePenalty      AwayPenalty      Result
## Min. :0.00000 Length :964 Length :964 Length :964
## 1st Qu.:0.00000 N.unique : 7 N.unique : 7 N.unique : 3
## Median :0.00000 N.blank : 0 N.blank : 0 N.blank : 0
## Mean :0.03631 Min.nchar: 1 Min.nchar: 1 Min.nchar: 4
## 3rd Qu.:0.00000 Max.nchar: 1 Max.nchar: 1 Max.nchar: 13
## Max. :1.00000
##
##      TournamentName      AwayScoreGroup
## 1998 FIFA World Cup: 64 Length :964
## 2002 FIFA World Cup: 64 N.unique : 3
## 2006 FIFA World Cup: 64 N.blank : 0
## 2010 FIFA World Cup: 64 Min.nchar: 8
## 2014 FIFA World Cup: 64 Max.nchar: 9
## 2018 FIFA World Cup: 64
## (Other) :580
```

```
# Check the AwayTeamScore groups.
```

```
table(matches_full$AwayScoreGroup)
```

```
##
```

##	0-1 goals	2-3 goals	4+ goals
##	701	237	26

Question 3 – Penalty Shootout Analysis

Objective

Filter out the matches that were decided by a penalty shootout. What are the top 5 national teams that most frequently participated in these matches? Do the rankings differ between home teams and away teams? Provide separate tables containing the columns TeamName, TeamCode, and NumberOfMatches, showing the top 5 home teams and top 5 away teams involved in penalty shootouts. Briefly discuss your findings.

Data Preparation

```
# PART A - Load required packages
#####

library(tidyverse)

#####
# PART B - Import and prepare the datasets
#####

# Read the Matches dataset.

matches <-
  read.csv(
    "data/Matches.csv",
    stringsAsFactors = FALSE
  )

# Read the Teams dataset.

teams <-
  read.csv(
    "data/Teams.csv",
    stringsAsFactors = FALSE
  )

#####
# PART C - Filter penalty shootout matches
#####

# Only keep matches decided by a penalty shootout.

penalty_matches <-
  matches %>%
  filter(
    PenaltyShootout == 1
  )

#####
# Check the filtered dataset
#####
```

```
# Display the first six rows.
```

```
nrow(penalty_matches)
```

```
## [1] 35
```

Top Five Home Teams

```
# PART D - Top 5 Home Teams
```

```
#####
```

```
# Merge penalty matches with the Teams dataset to obtain the home team name and code.
```

```
home_matches <-  
  penalty_matches %>%  
  left_join(  
    teams,  
    by = c("HomeTeamID" = "TeamID")  
  )
```

```
#####
```

```
# Check merged dataset
```

```
#####
```

```
# Display the first six rows.
```

```
head(home_matches)
```

```
##      MatchID TournamentID      MatchName      Stage  
## 1 M-1982-50      WC-1982    West Germany v France  semi-finals  
## 2 M-1986-45      WC-1986      Brazil v France quarter-finals  
## 3 M-1986-46      WC-1986    West Germany v Mexico quarter-finals  
## 4 M-1986-48      WC-1986      Spain v Belgium quarter-finals  
## 5 M-1990-41      WC-1990 Republic of Ireland v Romania  round of 16  
## 6 M-1990-45      WC-1990    Argentina v Yugoslavia quarter-finals  
##      MatchDate MatchTime StadiumID HomeTeamID AwayTeamID HomeTeamScore  
## 1 07/08/1982   21:00:00    S-159      T-83      T-28           3  
## 2  6/21/1986   12:00:00    S-095      T-09      T-28           1  
## 3  6/21/1986   16:00:00    S-102      T-83      T-44           0  
## 4  6/22/1986   16:00:00    S-104      T-71      T-06           1  
## 5  6/25/1990   17:00:00    S-073      T-58      T-59           0  
## 6  6/30/1990   17:00:00    S-071      T-03      T-84           0  
##      AwayTeamScore ExtraTime PenaltyShootout HomePenalty AwayPenalty      Result  
## 1           3           1           1           0           4 home team win  
## 2           1           1           1           0           4 away team win  
## 3           0           1           1           0           1 home team win  
## 4           1           1           1           0           5 away team win  
## 5           0           1           1           5           4 home team win  
## 6           0           1           1           3           2 home team win  
##      TeamName TeamCode
```

```
## 1      West Germany      DEU
## 2      Brazil            BRA
## 3      West Germany      DEU
## 4      Spain             ESP
## 5      Republic of Ireland IRL
## 6      Argentina         ARG
```

```
#####
# PART E - Identify the top five home teams
#####
```

```
#####
# Count the number of matches for each home team and sort the results in order.
#####
```

```
home_summary <-
  home_matches %>%
  group_by(
    TeamName,
    TeamCode
  ) %>%
  summarise(
    NumberOfMatches = n()
  ) %>%
  arrange(
    desc(NumberOfMatches)
  )
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by TeamName and TeamCode.
## i Output is grouped by TeamName.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(TeamName, TeamCode))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
#####
# Select the top five home teams
#####
```

```
top5_home <-
  head(
    home_summary,
    5
  )
```

```
#####
# Display the top five home teams
#####
```

```
top5_home
```

```
## # A tibble: 5 x 3
## # Groups:   TeamName [5]
```

##	TeamName	TeamCode	NumberOfMatches
##	<chr>	<chr>	<int>
## 1	Argentina	ARG	4
## 2	Brazil	BRA	4
## 3	Spain	ESP	4
## 4	Netherlands	NLD	3
## 5	West Germany	DEU	3

The results identify the **five home teams** that most frequently participated in FIFA World Cup matches decided by penalty shootouts. The analysis was selected by joining the **Matches** and **Teams** datasets to obtain team names, grouping the matches by team, counting the number of appearances, and sorting the results in descending order. These results indicate which teams have most often reached knockout matches that required penalty shootouts.

Top Five Away Teams

```
# PART F - Identify the top five away teams
#####

#####
# Merge penalty matches with the Teams dataset
#####

# Merge penalty matches with the Teams dataset to obtain the away team name and code.

away_matches <-
  penalty_matches %>%
  left_join(
    teams,
    by = c("AwayTeamID" = "TeamID")
  )

#####
# Check merged dataset
#####

# Display the first six rows.

head(away_matches)
```

##	MatchID	TournamentID	MatchName	Stage
## 1	M-1982-50	WC-1982	West Germany v France	semi-finals
## 2	M-1986-45	WC-1986	Brazil v France	quarter-finals
## 3	M-1986-46	WC-1986	West Germany v Mexico	quarter-finals
## 4	M-1986-48	WC-1986	Spain v Belgium	quarter-finals
## 5	M-1990-41	WC-1990	Republic of Ireland v Romania	round of 16
## 6	M-1990-45	WC-1990	Argentina v Yugoslavia	quarter-finals

##	MatchDate	MatchTime	StadiumID	HomeTeamID	AwayTeamID	HomeTeamScore
## 1	07/08/1982	21:00:00	S-159	T-83	T-28	3
## 2	6/21/1986	12:00:00	S-095	T-09	T-28	1
## 3	6/21/1986	16:00:00	S-102	T-83	T-44	0
## 4	6/22/1986	16:00:00	S-104	T-71	T-06	1
## 5	6/25/1990	17:00:00	S-073	T-58	T-59	0
## 6	6/30/1990	17:00:00	S-071	T-03	T-84	0

##	AwayTeamScore	ExtraTime	PenaltyShootout	HomePenalty	AwayPenalty	Result
## 1	3	1	1	0	4	home team win
## 2	1	1	1	0	4	away team win
## 3	0	1	1	0	1	home team win
## 4	1	1	1	0	5	away team win
## 5	0	1	1	5	4	home team win
## 6	0	1	1	3	2	home team win

##	TeamName	TeamCode
## 1	France	FRA
## 2	France	FRA
## 3	Mexico	MEX
## 4	Belgium	BEL
## 5	Romania	ROU

```
## 6 Yugoslavia      YUG
```

```
#####  
# Count the number of matches for each away team  
#####  
  
# Count the number of matches for each away team and sort the results in order.  
  
away_summary <-  
  away_matches %>%  
  group_by(  
    TeamName,  
    TeamCode  
  ) %>%  
  summarise(  
    NumberOfMatches = n()  
  ) %>%  
  arrange(  
    desc(NumberOfMatches)  
  )
```

```
## `summarise()` has regrouped the output.  
## i Summaries were computed grouped by TeamName and TeamCode.  
## i Output is grouped by TeamName.  
## i Use `summarise(groups = "drop_last")` to silence this message.  
## i Use `summarise(.by = c(TeamName, TeamCode))` for per-operation grouping  
##   (`?dplyr::dplyr_by`) instead.
```

```
#####  
# Select the top five away teams  
#####  
  
top5_away <-  
  head(  
    away_summary,  
    5  
  )  
  
#####  
# Display the top five away teams  
#####  
  
top5_away
```

```
## # A tibble: 5 x 3  
## # Groups:   TeamName [5]  
##   TeamName TeamCode NumberOfMatches  
##   <chr>    <chr>         <int>  
## 1 France   FRA             5  
## 2 Argentina ARG             3  
## 3 England  ENG             3  
## 4 Croatia  HRV             2  
## 5 Italy     ITA             2
```


Discussion

A similar procedure was applied to identify the **top five away teams** in penalty shootout matches. The findings show which teams most frequently appeared as the away team in these matches. Together with the home team analysis, the results provide a clearer understanding of the teams that have regularly competed in high-pressure knockout matches requiring penalty shootouts.

Validation

```
# PART G - Validate the results
#####

# Check the structure of the home team summary.

str(home_summary)

## gropd_df [20 x 3] (S3: grouped_df/tbl_df/tbl/data.frame)
## $ TeamName      : chr [1:20] "Argentina" "Brazil" "Spain" "Netherlands" ...
## $ TeamCode      : chr [1:20] "ARG" "BRA" "ESP" "NLD" ...
## $ NumberOfMatches: int [1:20] 4 4 4 3 3 2 2 1 1 1 ...
## - attr(*, "groups")= tibble [20 x 2] (S3: tbl_df/tbl/data.frame)
## ..$ TeamName: chr [1:20] "Argentina" "Brazil" "Colombia" "Costa Rica" ...
## ..$ .rows   : list<int> [1:20]
## .. ..$ : int 1
## .. ..$ : int 2
## .. ..$ : int 8
## .. ..$ : int 9
## .. ..$ : int 6
## .. ..$ : int 10
## .. ..$ : int 11
## .. ..$ : int 7
## .. ..$ : int 12
## .. ..$ : int 13
## .. ..$ : int 14
## .. ..$ : int 4
## .. ..$ : int 15
## .. ..$ : int 16
## .. ..$ : int 17
## .. ..$ : int 18
## .. ..$ : int 3
## .. ..$ : int 19
## .. ..$ : int 20
## .. ..$ : int 5
## .. ..@ ptype: int(0)
## ..- attr(*, ".drop")= logi TRUE

# Check the structure of the away team summary.

str(away_summary)

## gropd_df [25 x 3] (S3: grouped_df/tbl_df/tbl/data.frame)
```

```
## $ TeamName      : chr [1:25] "France" "Argentina" "England" "Croatia" ...
## $ TeamCode      : chr [1:25] "FRA" "ARG" "ENG" "HRV" ...
## $ NumberOfMatches: int [1:25] 5 3 3 2 2 1 1 1 1 1 ...
## - attr(*, "groups")= tibble [25 x 2] (S3: tbl_df/tbl/data.frame)
## ..$ TeamName: chr [1:25] "Argentina" "Belgium" "Brazil" "Bulgaria" ...
## ..$ .rows   : list<int> [1:25]
## .. ..$ : int 2
## .. ..$ : int 6
## .. ..$ : int 7
## .. ..$ : int 8
## .. ..$ : int 9
## .. ..$ : int 10
## .. ..$ : int 4
## .. ..$ : int 11
## .. ..$ : int 3
## .. ..$ : int 1
## .. ..$ : int 12
## .. ..$ : int 13
## .. ..$ : int 5
## .. ..$ : int 14
## .. ..$ : int 15
## .. ..$ : int 16
## .. ..$ : int 17
## .. ..$ : int 18
## .. ..$ : int 19
## .. ..$ : int 20
## .. ..$ : int 21
## .. ..$ : int 22
## .. ..$ : int 23
## .. ..$ : int 24
## .. ..$ : int 25
## .. ..@ ptype: int(0)
## ..- attr(*, ".drop")= logi TRUE
```

```
# Display the top five home teams.
```

```
top5_home
```

```
## # A tibble: 5 x 3
## # Groups:   TeamName [5]
##   TeamName      TeamCode NumberOfMatches
##   <chr>         <chr>           <int>
## 1 Argentina     ARG                 4
## 2 Brazil        BRA                 4
## 3 Spain         ESP                 4
## 4 Netherlands   NLD                 3
## 5 West Germany  DEU                 3
```

```
# Display the top five away teams.
```

```
top5_away
```

```
## # A tibble: 5 x 3
```

```
## # Groups:   TeamName [5]
##   TeamName TeamCode NumberOfMatches
##   <chr>    <chr>          <int>
## 1 France   FRA              5
## 2 Argentina ARG              3
## 3 England  ENG              3
## 4 Croatia  HRV              2
## 5 Italy    ITA              2
```

Question 4 – Factors Influencing Match Results

Objective

Write the code to analyse the factors that might influence the match outcome (Result) in the FIFA World Cup dataset. The Result variable consists of three categories: Home Team Win, Away Team Win, and Draw. Any factors in the dataset, such as Stage, ExtraTime, and Country, can be considered. Select two factors and investigate whether there are any trends that explain variations in match outcomes. Support your analysis with appropriate tables and visualisations, and briefly discuss your findings.

Data Presentation

```
# PART A - Load required packages
#####

library(tidyverse)

#####
# PART B - Import and prepare the dataset
#####
# Read the Matches dataset.
# This dataset contains match scores and match characteristics.

matches <-
  read.csv(
    "data/Matches.csv",
    stringsAsFactors = FALSE
  )

#####
# Check the dataset
#####

# Display the first six rows.

head(matches)
```

##	MatchID	TournamentID	MatchName	Stage	MatchDate
## 1	M-1930-01	WC-1930	France v Mexico	group stage	7/13/1930
## 2	M-1930-02	WC-1930	United States v Belgium	group stage	7/13/1930
## 3	M-1930-03	WC-1930	Yugoslavia v Brazil	group stage	7/14/1930
## 4	M-1930-04	WC-1930	Romania v Peru	group stage	7/14/1930
## 5	M-1930-05	WC-1930	Argentina v France	group stage	7/15/1930
## 6	M-1930-06	WC-1930	Chile v Mexico	group stage	7/16/1930

##	MatchTime	StadiumID	HomeTeamID	AwayTeamID	HomeTeamScore	AwayTeamScore
## 1	15:00:00	S-193	T-28	T-44	4	1
## 2	15:00:00	S-192	T-80	T-06	3	0
## 3	12:45:00	S-192	T-84	T-09	2	1
## 4	14:50:00	S-193	T-59	T-54	3	1
## 5	16:00:00	S-192	T-03	T-28	1	0

```
## 6 14:45:00 S-192 T-13 T-44 3 0
## ExtraTime PenaltyShootout HomePenalty AwayPenalty Result
## 1 0 0 ? 0 home team win
## 2 0 0 0 0 home team win
## 3 0 0 0 0 home team win
## 4 0 0 0 ? home team win
## 5 0 0 0 0 home team win
## 6 0 0 0 0 home team win
```

#Display the structure of the dataset.

```
str(matches)
```

```
## 'data.frame': 964 obs. of 16 variables:
## $ MatchID : chr "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName : chr "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v" ...
## $ Stage : chr "group stage" "group stage" "group stage" "group stage" ...
## $ MatchDate : chr "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime : chr "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
## $ StadiumID : chr "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID : chr "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID : chr "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore : int 4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore : int 1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime : int 0 0 0 0 0 0 0 0 0 0 ...
## $ PenaltyShootout: int 0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty : chr "?" "0" "0" "0" ...
## $ AwayPenalty : chr "0" "0" "0" "?" ...
## $ Result : chr "home team win" "home team win" "home team win" "home team win" ...
```

```
#####
# PART C - Prepare the dataset
#####
```

Convert Stage into a factor. Stage is a categorical variable.

```
matches$Stage <-
  factor(
    matches$Stage,
    levels = c(
      "group stage",
      "second group stage",
      "round of 16",
      "quarter-finals",
      "semi-finals",
      "third-place match",
      "final",
      "final round"
    )
  )
```

Convert ExtraTime into a factor. ExtraTime contains two categories (0 and 1).

```

matches$ExtraTime <-
  factor(matches$ExtraTime)

# Convert Result into a factor. Result contains the match outcome categories.

matches$Result <-
  factor(matches$Result)

#####
# Check the updated data types
#####

# Display the structure of the dataset.

str(matches)

```

```

## 'data.frame':    964 obs. of  16 variables:
## $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName    : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v" ...
## $ Stage        : Factor w/ 8 levels "group stage",...: 1 1 1 1 1 1 1 1 1 ...
## $ MatchDate    : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime    : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
## $ StadiumID    : chr  "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID   : chr  "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID   : chr  "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore : int  4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore : int  1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime    : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 ...
## $ PenaltyShootout: int  0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty   : chr  "?" "0" "0" "0" ...
## $ AwayPenalty   : chr  "0" "0" "0" "?" ...
## $ Result       : Factor w/ 3 levels "away team win",...: 3 3 3 3 3 3 3 3 3 ...

```

Match Results by Competition Stage

```
# PART D - Explore match results by Stage
#####

# Check Stage values
#####

# Display the number of matches in each competition stage.

table(matches$Stage)
```

```
##
##      group stage second group stage      round of 16      quarter-finals
##      676          36          97          70
##      semi-finals third-place match      final      final round
##      38          20          21          6
```

```
#####
# PART E - Create the first visualisation
#####

# Create a bar chart of match results by competition stage.
# A bar chart is used because Stage and Result are categorical variables.
# The chart clearly compares the number of match results across competition stages.
```

```
ggplot(
  matches,
  aes(
    x = Stage,
    fill = Result
  )
) +
  geom_bar(
    position = "dodge"
  ) +
  labs(
    title = "Match Results by Competition Stage",
    x = "Competition Stage",
    y = "Number of Matches",
    fill = "Match Result"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    )
  ) +
  scale_x_discrete(
    labels = c(
```

```

    "final" = "Final",
    "final round" = "Final Round",
    "group stage" = "Group Stage",
    "quarter-finals" = "Quarter-finals",
    "round of 16" = "Round of 16",
    "second group stage" = "Second Group Stage",
    "semi-finals" = "Semi-finals",
    "third-place match" = "Third-place Match"
  )
) +
scale_fill_discrete(
  labels = c(
    "away team win" = "Away Team Win",
    "draw" = "Draw",
    "home team win" = "Home Team Win"
  )
)
)

```

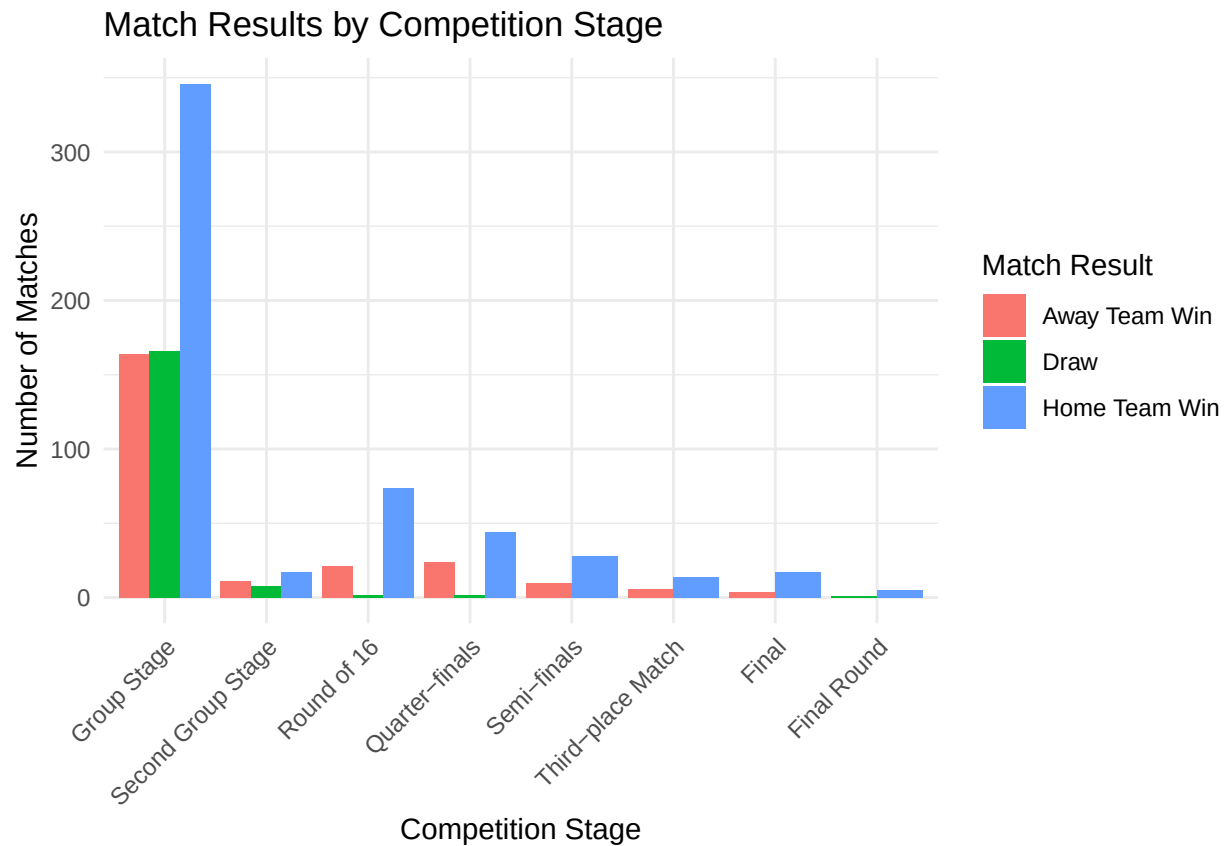


Figure 5. Match Results by Competition Stage.

Discussion

The bar chart shows the number of **home team wins**, **away team wins**, and **draws** at each stage of the FIFA World Cup. The **group stage** has the **highest** number of matches because it includes the most games in the tournament. The number of matches then decreases in the knockout rounds, leading to fewer results at each later stage. Overall, home team wins are **more common** than away team wins in most stages, while draws are **less frequent** since knockout matches must end with a winner.

Match Results by Extra Time

```
# PART F - Create the second visualisation
#####

# Create a bar chart of match results by extra time.
# A bar chart is used because ExtraTime and Result are categorical variables.
# The chart compares match results between matches with and without extra time.

ggplot(
  matches,
  aes(
    x = ExtraTime,
    fill = Result
  )
) +
  geom_bar(
    position = "dodge"
  ) +
  labs (
    title = "Match Results by Extra Time",
    x = "Extra Time",
    y = "Number of Matches",
    fill = "Match Result"
  ) +
  theme_minimal() +
  scale_x_discrete(
    labels = c(
      "0" = "No",
      "1" = "Yes"
    )
  ) +
  scale_fill_discrete(
    labels = c(
      "away team win" = "Away Team Win",
      "draw" = "Draw",
      "home team win" = "Home Team Win"
    )
  )
)
```

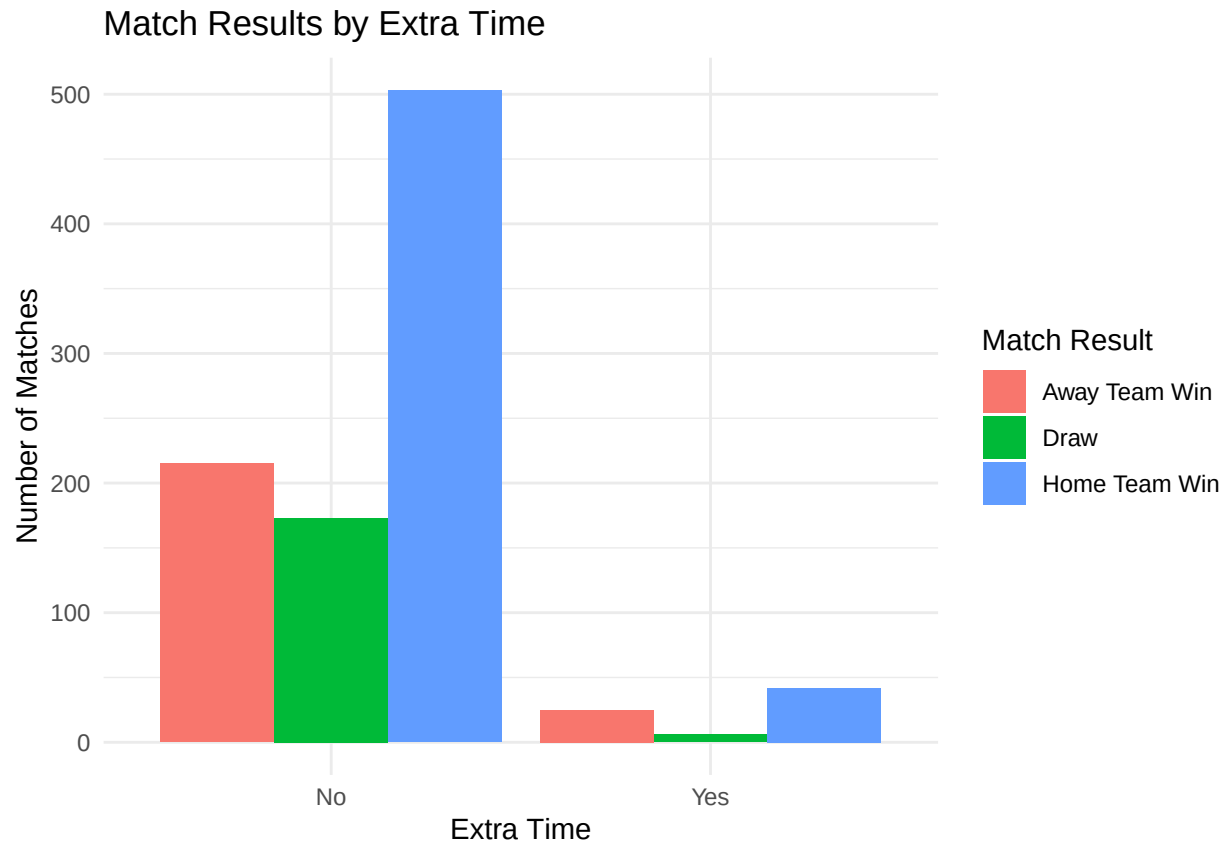


Figure 6. Match Results by Extra Time.

Discussion

The second bar chart compares match results between games that **did not require extra time** and those that **went to extra time**. Most FIFA World Cup matches were decided within regular time, so the **No** category contains far more matches than the **Yes** category. Across both groups, **home team wins** were the most common outcome. **Draws** were more frequent in matches that went to extra time because these games were tied at the end of regular time.

Validation

```
# PART G - Validate the results
#####

# Display the structure of the dataset.

str(matches)
```

```
## 'data.frame':   964 obs. of  16 variables:
## $ MatchID      : chr  "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr  "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName    : chr  "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "Romania v ..."
## $ Stage        : Factor w/ 8 levels "group stage",...: 1 1 1 1 1 1 1 1 1 1 ...
```

```
## $ MatchDate      : chr  "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime      : chr  "15:00:00" "15:00:00" "12:45:00" "14:50:00" ...
## $ StadiumID      : chr  "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID      : chr  "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID      : chr  "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore   : int   4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore   : int   1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime       : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
## $ PenaltyShootout: int   0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty      : chr  "?" "0" "0" "0" ...
## $ AwayPenalty      : chr  "0" "0" "0" "?" ...
## $ Result           : Factor w/ 3 levels "away team win",...: 3 3 3 3 3 3 3 3 3 3 ...
```

Display the number of matches in each competition stage.

```
table(matches$Stage)
```

```
##
##      group stage second group stage      round of 16      quarter-finals
##           676           36           97           70
##      semi-finals third-place match      final      final round
##           38           20           21           6
```

Display the number of matches with and without extra time.

```
table(matches$ExtraTime)
```

```
##
##      0      1
## 891  73
```

Display the number of each match result.

```
table(matches$Result)
```

```
##
## away team win      draw home team win
##           240           179           545
```

GitHub Repository

Objective

Git and **GitHub** were used throughout this assignment to manage *version control* and *document* the project's development. **Git** was used to track changes made to the project files, while **GitHub** served as the remote repository for storing and backing up the project. This section outlines the Git commands used, presents the commit history, and provides the link to the GitHub repository.

GitHub Repository Link

GitHub Repository: https://github.com/24007292-VuKim/COMP1013_Project

Git Command

Git Command	Explanation
<code>git init</code>	Creates a new local Git repository.
<code>git status</code>	Displays the status of files in the repository.
<code>git add .</code>	Stages all modified files for the next commit.
<code>git commit -m "message"</code>	Saves the staged changes with a descriptive commit message.
<code>git remote add origin <repository-url></code>	Connects the local repository to the remote GitHub repository.
<code>git push -u origin main</code>	Uploads the commits from the local repository to GitHub.

Commit History

Figure 7 shows the commit history recorded during the development of this assignment. Multiple commits were made throughout the project to document progress, track changes, and maintain version control during the analysis.

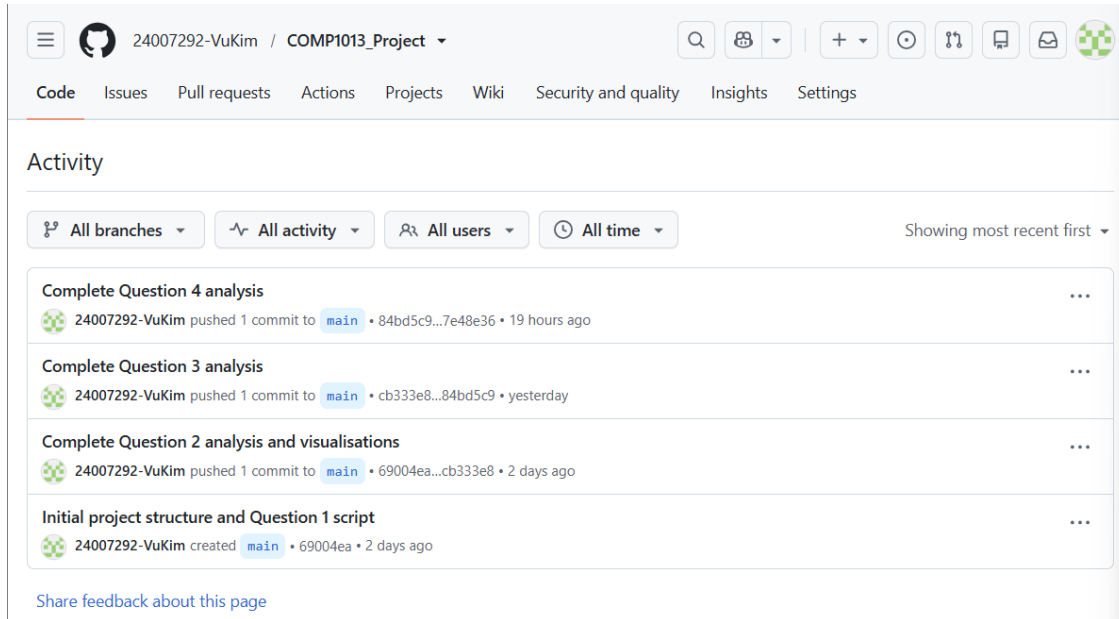


Figure 7. Git Commit History

Conclusion

This report demonstrated the use of **R** and the **tidyverse** package to analyse FIFA World Cup data. The project followed the complete data analysis process, including importing, inspecting, and cleaning the data, transforming variables, creating visualisations, and interpreting the findings. Statistical graphs were used to explore match outcomes, competition stages, extra time, penalty shootouts, and team performance. In addition, **Git** and **GitHub** were used for version control, enabling the project's development to be tracked through meaningful commits. Overall, this assignment provided practices in data analysis, data visualisation, report performing with **R Markdown**, and project management using Git, strengthening the key skills developed throughout the COMP1013 course.